Healthcare Resources and Service Use Across China's Provinces, 2011–2023 Final Project Research Report

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1. Introduction

China has expanded its health system rapidly, but resources are not spread evenly, and more resources do not automatically mean proportionally more use. That matters for policy: hospital beds are expensive, and they only improve access if people actually use the capacity. This report asks: how does healthcare resource availability (health personnel, hospital beds, and facilities) vary across Chinese provinces, and how is it associated with the utilization of healthcare services (outpatient visits, inpatient discharges, and bed utilization rate)? I approach it from economics and policy research, interested in how regional development and policy choices relate to access and use.

2. Data and Methods

I use provincial health statistics from the National Bureau of Statistics of China[1] for 31 provinces from 2011 to 2023 (403 province-year observations). I dropped 2024 because bed utilization is missing for every province, and Several indicators labelled "per 1,000 population" behave like totals, so I treated them as totals and merged in provincial population to build true per-capita measures: beds and personnel per 1,000 people, institutions per 10,000 people, discharges per 1,000 people, and visits per person. I grouped provinces into East, Central, West, and Northeast regions.

The analysis has three steps. First, descriptive statistics and regional comparisons for 2019 (a pre-COVID baseline) and 2023. Second, Pearson and Spearman correlations[2] between resources and utilization. Third, an OLS regression[3] on the 2019 cross-section: discharges per 1,000 people on beds per 1,000, personnel per 1,000, and institutions per 10,000. I checked residuals,

[1] National Bureau of Statistics of China, provincial health statistics (health personnel, hospital beds, health care institutions, outpatient visits, patients discharged, hospital bed utilization rate), data.stats.gov.cn, 2011–2023. Provincial population (10,000 persons): [https://www.stats.gov.cn/english/]

[2] Pearson, K. (1895). Notes on regression and inheritance in the case of two parents. Proceedings of the Royal Society of London, 58, 240–242. Spearman, C. (1904). The proof and measurement of association between two things. American Journal of Psychology, 15(1), 72–101.

[3] Wooldridge, J. M. (2019). Introductory Econometrics: A Modern Approach (7th ed.). Cengage. Models fit with statsmodels: Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with Python. Proceedings of the 9th Python in Science Conference.

variance inflation factors, heteroskedasticity-robust (HC3) standard errors[4], and influential observations, and repeated the model for 2023 and without Xizang.

3. Descriptive Findings

Resources grew in every region. The provincial mean of beds per 1,000 people rose from 3.77 in 2011 to 6.38 in 2019 and 7.48 in 2023, and personnel per 1,000 rose from 9.67 to 11.71 between 2019 and 2023 (Table 1, Figure 1). In 2019, beds ranged from 4.67 (Guangdong) to 7.82 (Xinjiang), and personnel from 7.56 to 15.69, with Beijing a clear outlier. Surprisingly, the wealthier East has the fewest beds per person (5.59 in 2019, versus 6.81 in the West and 7.06 in the Northeast; Figure 2).

Service use shows a different geography. East provinces average 8.22 outpatient visits per person, roughly double the Northeast (4.01), led by Beijing, Zhejiang and Shanghai at over 11. Inpatient discharges are instead highest in the West (201 per 1,000) and Central region (199), and lowest in the East (164) and Northeast (163). Bed utilization fell from 81.98% in 2019 to 77.72% in 2023, with the provincial mean dropping to 70.3% in 2020 and 68.5% in 2022 before a partial recovery (Figure 1). Inequality in discharges also widened: the coefficient of variation rose from 0.21 to 0.23.

Table 1. Descriptive statistics across 31 provinces

Variable	2019 mean (SD)	2019 range	2023 mean (SD)	2023 range
Beds per 1,000	6.38 (0.88)	4.67-7.82	7.48 (0.85)	5.41-8.93
Personnel per 1,000	9.67 (1.44)	7.56-15.69	11.71 (1.89)	8.88-19.22
Institutions per 10,000	7.77 (3.44)	2.28-21.03	8.41 (3.46)	2.76-22.84
Discharges per 1,000	184.99 (38.76)	92.33-246.34	216.51 (49.12)	114.47-324.01
Visits per person	6.13 (2.08)	3.20-11.38	6.88 (2.41)	3.17-13.59
Bed utilization (%)	81.98 (6.45)	64.80-96.20	77.72 (7.15)	54.40-91.00

Figure 1. Provincial mean, 2011-2023 (shaded = 2020-2022)

[4] MacKinnon, J. G., & White, H. (1985). Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. Journal of Econometrics, 29(3), 305–325. Cook's distance: Cook, R. D. (1977). Detection of influential observation in linear regression. Technometrics, 19(1), 15–18.

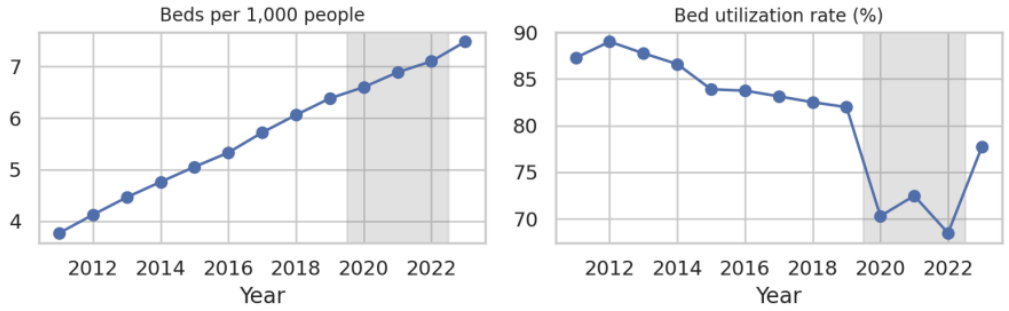
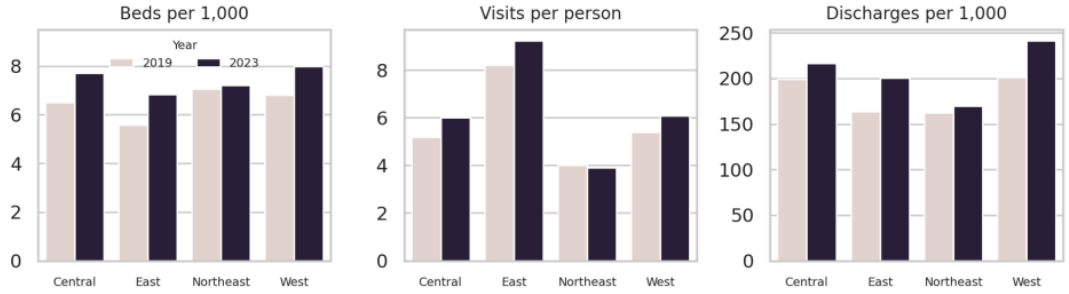


Figure 2. Regional means, 2019 vs 2023



4. Correlation and Regression Results

In 2019, beds per 1,000 are strongly correlated with discharges per 1,000 (Pearson r = 0.72, Spearman 0.64) but negatively with visits per person (r = −0.42). Personnel are mildly positively correlated with visits (r = 0.44), and institutions per 10,000 are negatively correlated with bed utilization (r = −0.60). The regression (Table 2) explains 63.3% of the variation in discharges (adjusted R2 = 0.59; F-test p < 0.001; n = 31). Holding the other resources constant, one additional bed per 1,000 people is associated with about 33 more discharges per 1,000 (95% CI 22.7 to 43.8). Personnel show no meaningful association (p = 0.70). Institutions are negatively associated: each additional institution per 10,000 people goes with 3.7 fewer discharges (p = 0.010). My interpretation is that provinces with many small clinics spread over sparse populations, such as Xizang, have fewer inpatient stays per person. Diagnostics were reassuring: VIFs were about 1.0, the Breusch-Pagan test showed no heteroskedasticity (p = 0.39), residuals were approximately normal (Jarque-Bera p = 0.31), and no province was highly influential (maximum Cook's D = 0.19). HC3 standard errors leave conclusions unchanged. Dropping Xizang leaves the beds coefficient almost identical (33.0) but the institutions estimate (−3.6) becomes imprecise (p = 0.11), so that finding is less secure. In 2023 the model fits better (R2 = 0.81) and the beds coefficient is larger (48.3), suggesting the link between capacity and use strengthened after COVID-19.

Figure 3. Beds vs discharges, 2019 (31 provinces)

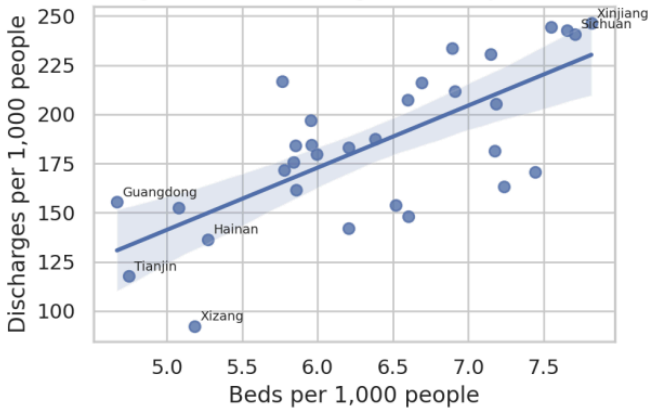


Table 2. OLS regression, 2019: discharges per 1,000 people (n = 31, R^2 = 0.633)

Predictor	Coefficient	Std. error	p-value	95% CI
Intercept	13.62	44.99	0.764	-78.70, 105.94
Beds per 1,000	33.25	5.15	<0.001	22.67, 43.82
Personnel per 1,000	-1.25	3.19	0.698	-7.80, 5.30
Institutions per 10,000	-3.70	1.34	0.010	-6.46, -0.94

5. Conclusions and Limitations

Bed supply closely tracks inpatient use across provinces, but outpatient use is concentrated in the East, where beds per person are lowest. This suggests visits are driven by factors beyond beds, such as income, urbanization, or primary-care access, so policymakers should not rely on bed counts alone to judge access. These results are associations, not causal effects. Provinces with more resources may have older or wealthier populations that use more care (confounding), and I did not include GDP per capita. Provincial patterns may not hold for individuals (ecological fallacy).

With only 31 provinces, estimates are imprecise, and COVID-19 distorts utilization in 2020–2022. Beds and discharges are also mechanically related, since beds are the capacity through which discharges occur. Regional means are unweighted, and per-capita measures rely on my assumption about indicator units.